

Gradient-Based Leakage Attacks on GCNNs: A Powerful Threat for Secure Circuit Design

Appendix

A. Gradient Clipping and Perturbation

Here, we implement a defense mechanism that combines gradient clipping with perturbation. Let

$$G_{\text{true}}^{(l)} = \nabla_{W^{(l)}} \mathcal{L}(\hat{Y}, Y)$$

denote the true gradient of the loss $\mathcal{L}$ with respect to the model parameters $W^{(l)}$ at layer l . We first clip these gradients to restrict their magnitudes by enforcing

$$G_{\text{clip}}^{(l)} = \min \left(1, \frac{C}{\|G_{\text{true}}^{(l)}\|_2} \right) G_{\text{true}}^{(l)},$$

where the clipping threshold is set to $C = 1.0$. To further obscure the leaked information, we add Gaussian noise to produce the defended gradient:

$$G_{\text{def}}^{(l)} = G_{\text{clip}}^{(l)} + \mathcal{N}(0, \alpha^2 I),$$

with $\alpha = 0.05$ and I being the identity matrix. The reconstruction objective then becomes

$$\min_{\tilde{X}} \sum_{l=0}^{L-1} \left\| \nabla_{W^{(l)}} \mathcal{L}(\tilde{X}) - G_{\text{def}}^{(l)} \right\|_F^2,$$

where $\tilde{X}$ is the dummy input being optimized. This two-fold defense – clipping gradients to a maximum norm of 1.0 and perturbing them with a noise multiplier of 0.05 – effectively degrades the precision of the gradient signal available to the attacker, leading to higher reconstruction errors and improved privacy protection.

Figure 1 shows that applying gradient clipping and perturbation increases the reconstruction errors across all gate types compared to the attack without any prevention technique.

B. Secure Aggregation Protocols

Here, we apply secure aggregation protocols, wherein gradients are computed and averaged over multiple nodes instead of individual samples. This aggregated approach limits the attacker’s ability to reconstruct specific node features while preserving the integrity of model updates.

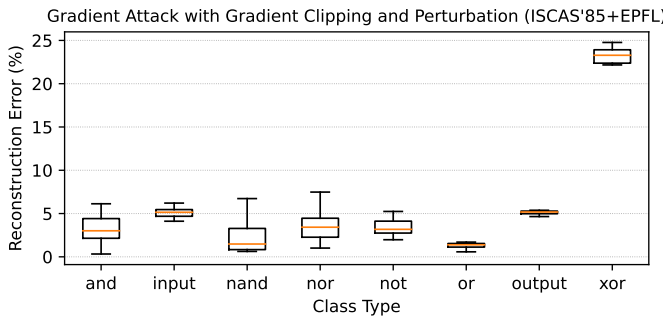


Fig. 1: Reconstruction errors under gradient clipping and perturbation.

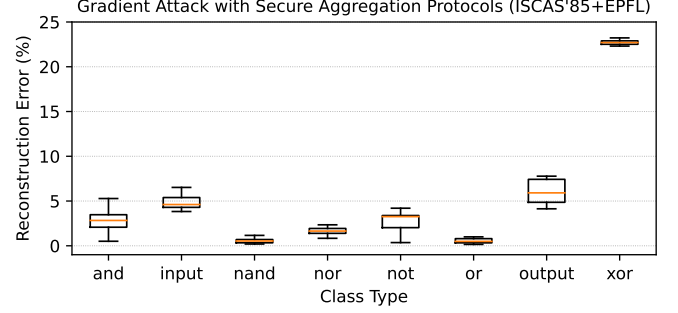


Fig. 2: Reconstruction errors under secure aggregation protocols.

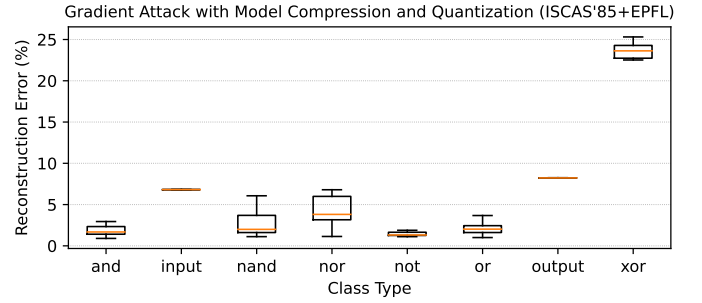


Fig. 3: Reconstruction errors under secure aggregation protocols.

Let a group of N_{group} nodes be selected from a given class, each with its corresponding feature X_i . The aggregated feature for the group is computed as:

$$X_{\text{agg}} = \frac{1}{N_{\text{group}}} \sum_{i=1}^{N_{\text{group}}} X_i.$$

Similarly, the aggregated gradients for the group are computed using the mean loss over all selected nodes:

$$G_{\text{agg}}^{(l)} = \frac{1}{N_{\text{group}}} \sum_{i=1}^{N_{\text{group}}} \nabla_{W^{(l)}} \mathcal{L}(X_i, Y_i),$$

where $\mathcal{L}(X_i, Y_i)$ represents the loss function for node i .

In our implementation, we set $N_{\text{group}} = 5$ and simulate an attacker attempting to reconstruct X_{agg} rather than individual node features. The reconstruction objective now becomes:

$$\min_{\tilde{X}_{\text{agg}}} \sum_{l=0}^{L-1} \left\| \nabla_{W^{(l)}} \mathcal{L}(\tilde{X}_{\text{agg}}) - G_{\text{agg}}^{(l)} \right\|_F^2.$$

Since the attacker is only provided with the averaged gradient and feature information, individual variations within the group are masked, notably degrading inversion performance, as shown in Figure 2.

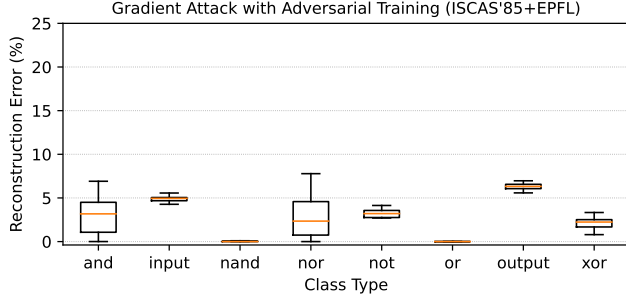


Fig. 4: Reconstruction errors under adversarial training.

C. Model Compression and Quantization

Here, we use model compression and quantization techniques to restrict the precision of model parameters, obfuscating critical information that could be exploited by an attacker.

1) *Quantization*: During the forward pass of the GCNN, numerical outputs are quantized using:

$$X_{\text{quant}} = \frac{\text{round}(X \cdot Q)}{Q},$$

where $Q = 255$ simulates 8-bit fixed-point representation. This transformation limits the granularity of feature updates, reducing inversion fidelity.

2) *Compression via Pruning*: After backpropagation, small-magnitude model weights are set to zero:

$$W_{\text{compressed}}^{(l)} = \begin{cases} 0, & \text{if } |W^{(l)}| < \tau \\ W^{(l)}, & \text{otherwise} \end{cases}$$

where $\tau = 0.01$ is the pruning threshold. This process eliminates weak connections in the NN, making gradient inversion less effective.

Figure 3 illustrates that model compression and quantization can indeed increase reconstruction errors compared to an unprotected setting.

D. Adversarial Training

Here, we introduce adversarial perturbations into the model's training process to improve its resilience against GBLAs. By exposing the model to carefully crafted perturbations during training, it learns representations that are less susceptible to inversion attacks.

1) *Generating Adversarial Perturbations*: To generate adversarial examples, we use the Fast Gradient Sign Method (FGSM), where the perturbation applied to the feature matrix X is computed as:

$$X_{\text{adv}} = X + \epsilon \cdot \text{sign}(\nabla_X \mathcal{L}(X, Y)),$$

where ϵ is the perturbation strength (set to 0.1 in our experiments), $\mathcal{L}(X, Y)$ is the loss function, and $\nabla_X \mathcal{L}$ represents the gradient of the loss with respect to the input features.

2) *Adversarial Training Procedure*: During training, instead of using the original features, we compute adversarial perturbations and feed the model with X_{adv} , reinforcing its ability to generalize across perturbed data:

$$\min_W \sum_{i=1}^N \mathcal{L}(f_W(X_{\text{adv},i}), Y_i).$$

This forces the model to optimize for robustness rather than merely fitting the clean training data.

3) *Impact on Reconstruction Error*: Figure 4 illustrates the reconstruction errors observed when adversarial training is applied as a defense against GBLAs.